
Air Quality Level Classification System

Project Id: AIML-19

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1. Problem Description	3
2. Architecture	4
3. Models	5
4. Evaluation metrics used	6
5. Preliminary Results and Visualizations.....	7
6. References.....	9

1. Problem Description:

The Air Quality Level Classification System is a Python-based machine learning application developed to analyse environmental sensor data and classify air quality into predefined categories such as Good, Moderate, Poor, or Hazardous. It simplifies the process of collecting, processing, and interpreting air pollution data from various sources, ensuring accurate and efficient assessment of air conditions.

Using core concepts of Machine Learning such as data pre-processing, feature engineering, classification algorithms, model evaluation, and data visualization, this system identifies patterns in pollutant levels like CO, NO₂, PM2.5, and PM10. The model helps in predicting air quality levels in real-time, reducing manual monitoring efforts and enabling early detection of harmful conditions.

This project supports informed decision-making for public health, environmental monitoring, and urban planning by providing reliable insights into air quality trends and potential risks.

2. Architecture:

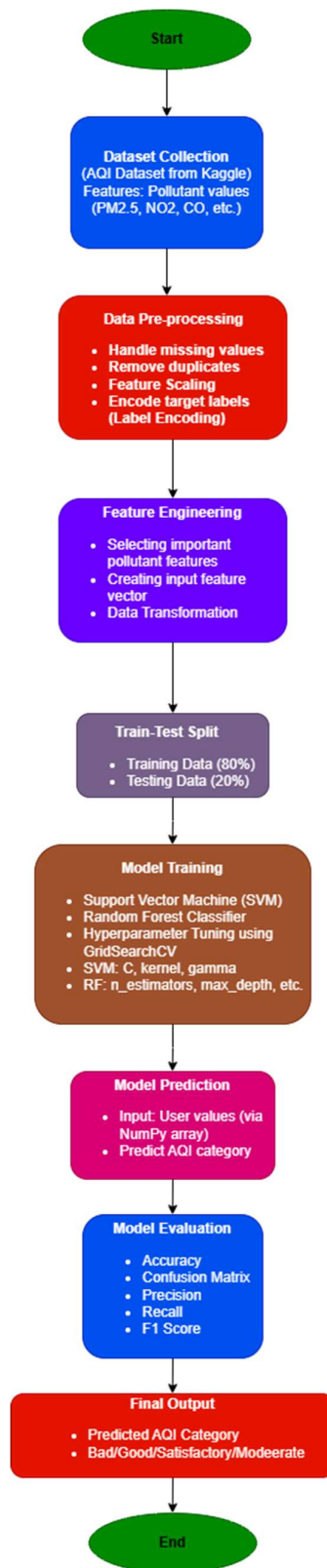


Fig.1 Sample flow diagram.

3. Models Used:

1. Logistic Regression (Primary Model)

Logistic Regression is the main model used for classification of Air Quality Index (AQI) categories. It is a simple, efficient, and widely used supervised learning algorithm that predicts the probability of a data point belonging to a particular class. It performs well when the data is relatively linearly separable and provides easily interpretable results.

Hyperparameters tuned:

- C- Regularization strength
- penalty- L1 / L2 regularization
- solver- Optimization algorithm

2. Support Vector Machine (SVM)

Support Vector Machine is used as a comparative model for AQI classification. It works by finding the optimal hyperplane that separates different classes with maximum margin.

Hyperparameters tuned:

- C- Controls regularization
- Kernel- Linear / RBF
- Gamma- Influence of data points

3. Decision Tree Classifier

Decision Tree is a supervised learning algorithm that classifies data by splitting it into branches based on feature values.

Hyperparameters tuned:

- max_depth- Maximum depth of the tree
- min_samples_split- Minimum samples required to split a node
- min_samples_leaf- Minimum samples required at a leaf node

4. Random Forest Classifier

Random Forest is an ensemble learning technique that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting.

Hyperparameters tuned:

- n_estimators- Number of trees
- max_depth- Depth of trees
- min_samples_split,
- min_samples_leaf

5. K-Nearest Neighbors (KNN)

KNN is a simple classification algorithm that assigns class labels based on the majority class among the nearest data points.

k- Number of nearest neighbors

4. Evaluation metrics used:

To evaluate the performance of the machine learning models in the Air Quality Level Classification System, several standard classification metrics were used. These metrics help in assessing how accurately the model predicts different air quality categories.

1. Accuracy

Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total instances.

- Helps in understanding the general performance of the model
- Best used when the dataset is balanced

2. Precision

Precision measures how many of the predicted positive instances are actually correct.

- Important when false positives need to be minimized
- Indicates reliability of positive predictions

3. Recall (Sensitivity)

Recall measures how many actual positive instances were correctly identified by the model.

- Important when missing a positive case is costly
- Helps in identifying how well the model detects critical cases (e.g., poor air quality)

4. F1-Score

F1-score is the harmonic mean of Precision and Recall, providing a balance between the two.

- Useful when there is an imbalance in class distribution
- Gives a single performance measure combining precision and recall

5. Confusion Matrix

A confusion matrix is used to visualize the performance of the classification model by showing actual vs predicted values.

- Helps in identifying types of errors (false positives and false negatives)
- Useful for detailed performance analysis

Conclusion

All models were evaluated using the above metrics, and their results were compared. The model with the best balance of accuracy, precision, recall, and F1-score was selected as the final model for predicting air quality levels.

5. Preliminary Results and Visualizations

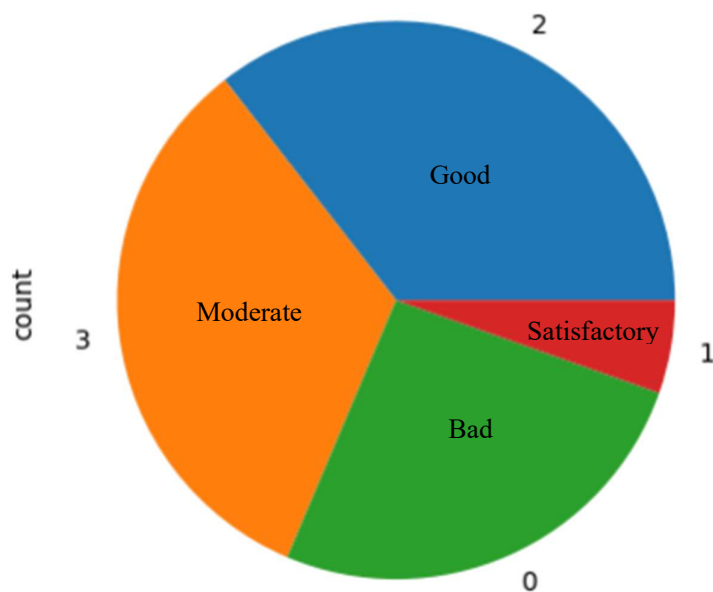


Fig.2 AQI Category Distribution

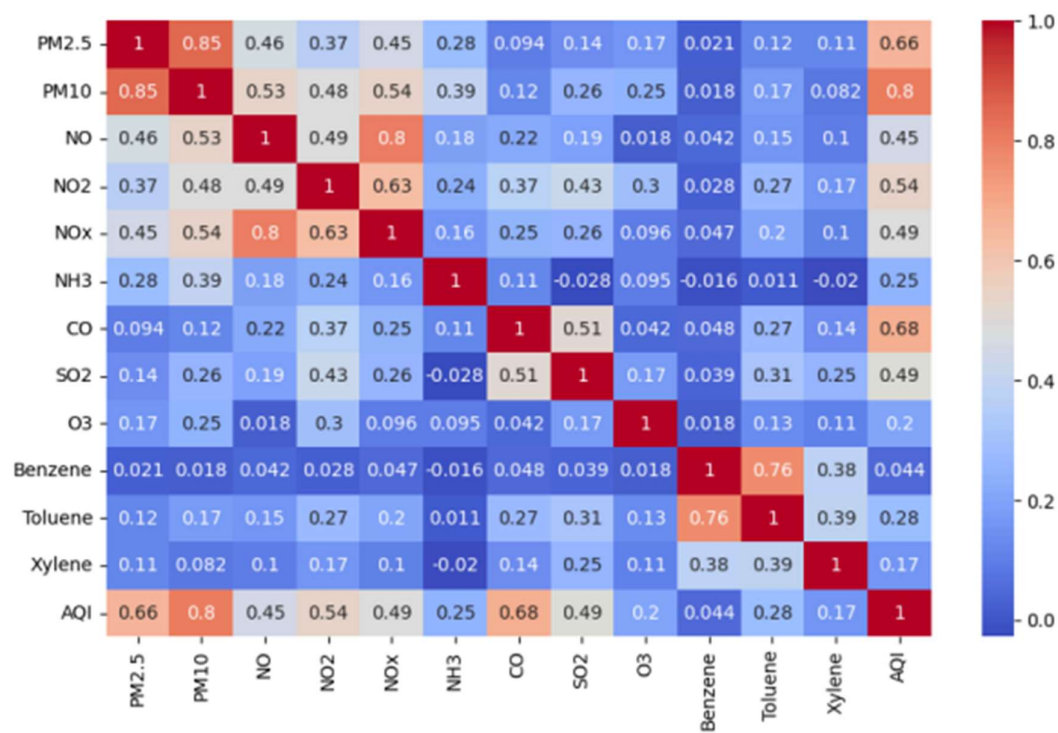


Fig.3 Heatmap showing correlation.

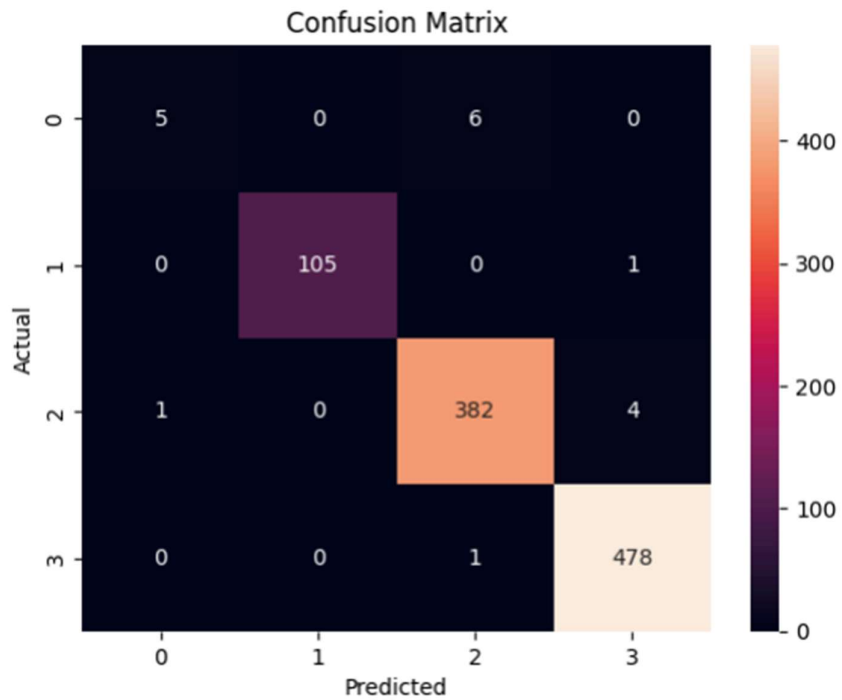


Fig.4 Logistic Regression Confusion Matrix

	precision	recall	f1-score	support
0	0.83	0.45	0.59	11
1	1.00	0.99	1.00	106
2	0.98	0.99	0.98	387
3	0.99	1.00	0.99	479
accuracy			0.99	983
macro avg	0.95	0.86	0.89	983
weighted avg	0.99	0.99	0.99	983

Fig.5 Logistic Regression Classification Report

RESULT (Logistic Regression):

```
Input: 0, 0, 0, 0, 13, 9, 0, 1, 3, 0, 0, 0, 0
Output:
RAW PREDICTION [3]
Air Quality: Satisfactory
```

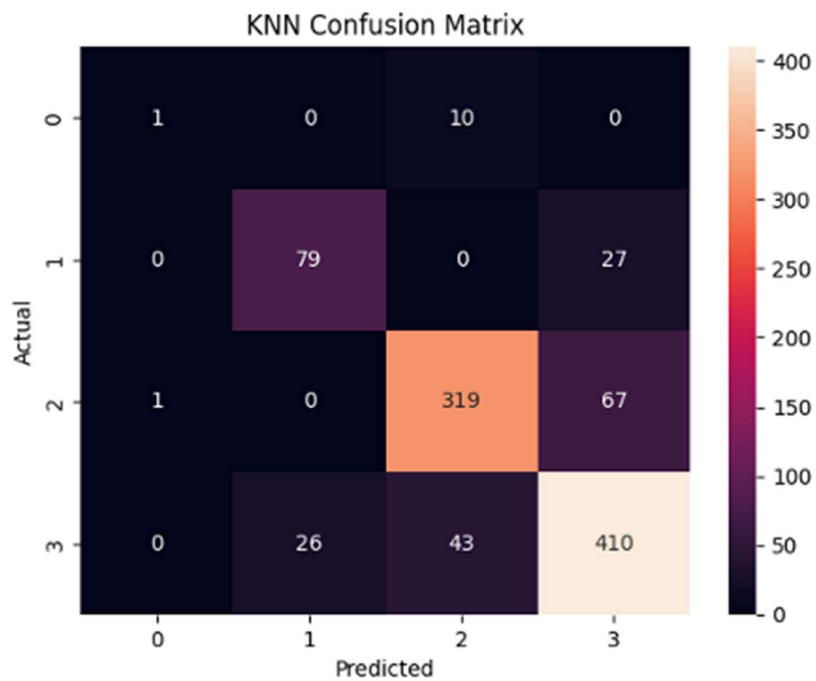



Fig.6 KNN Regression Confusion Matrix

	precision	recall	f1-score	support
0	0.50	0.09	0.15	11
1	0.75	0.75	0.75	106
2	0.86	0.82	0.84	387
3	0.81	0.86	0.83	479
accuracy			0.82	983
macro avg	0.73	0.63	0.64	983
weighted avg	0.82	0.82	0.82	983

Fig.7 KNN Classification Report

RESULT (KNN):

```

Input: 4.38, 74.09, 3.42, 26.06, 16.53, 11.99, 0.52, 12.72, 30.14, 0.74, 2.21, 0.38,12
Output:
RAW PREDICTION [1]
Air Quality: Moderate

```

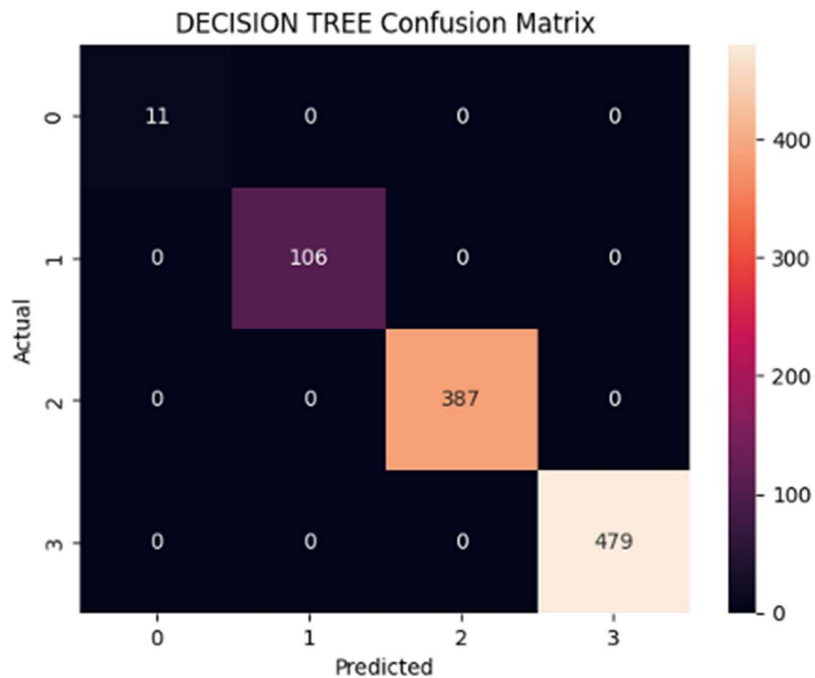


Fig.8 Decision Tree Confusion Matrix

	precision	recall	f1-score	support
0	1.00	1.00	1.00	11
1	1.00	1.00	1.00	106
2	1.00	1.00	1.00	387
3	1.00	1.00	1.00	479
accuracy			1.00	983
macro avg	1.00	1.00	1.00	983
weighted avg	1.00	1.00	1.00	983

Fig.9 Decision Tree Classification Report

RESULT (Decision Tree):

```

Input: 4.38, 74.09, 3.42, 26.06, 16.53, 11.99, 0.52, 12.72, 30.14, 0.74, 2.21, 0.38,12
Output:
RAW PREDICTION [2]
Air Quality: Poor

```

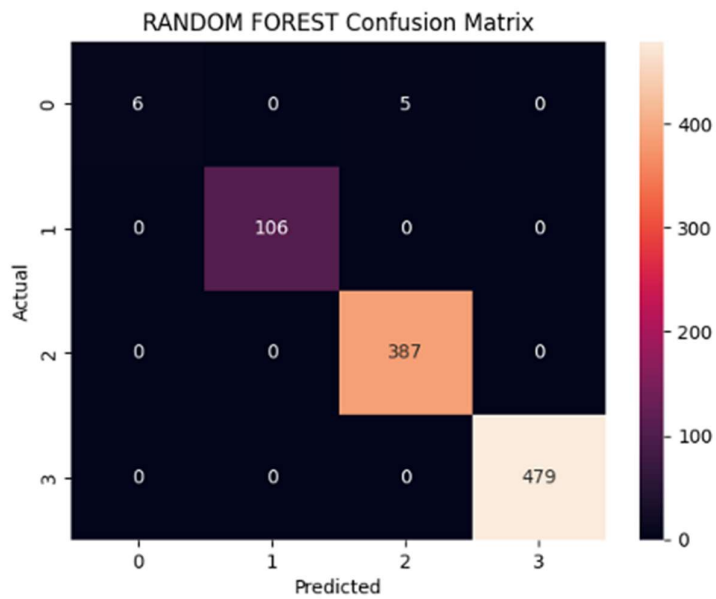


Fig.10 Random Forest Confusion Matrix

	precision	recall	f1-score	support
0	1.00	0.55	0.71	11
1	1.00	1.00	1.00	106
2	0.99	1.00	0.99	387
3	1.00	1.00	1.00	479
accuracy			0.99	983
macro avg	1.00	0.89	0.92	983
weighted avg	0.99	0.99	0.99	983

Fig.11 Random Forest Classification Report

RESULT (Random Forest):

```
Input: 0, 0, 0, 0, 13, 9, 0, 1, 3, 0, 0, 0, 0
Output:
RAW PREDICTION [3]
Air Quality: Satisfactory
```

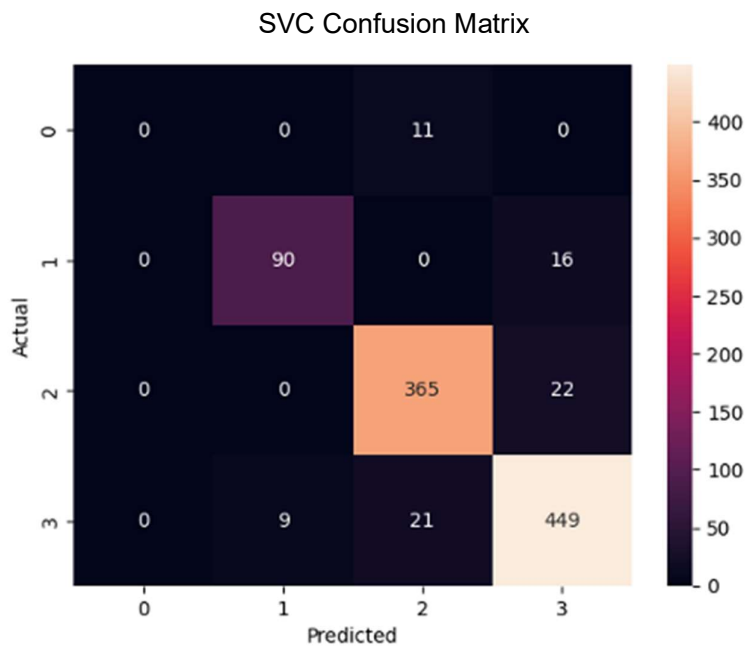


Fig.12 SVC Confusion Matrix

	0	0.00	0.00	0.00	11
	1	0.91	0.85	0.88	106
	2	0.92	0.94	0.93	387
	3	0.92	0.94	0.93	479
accuracy				0.92	983
macro avg		0.69	0.68	0.68	983
weighted avg		0.91	0.92	0.91	983

Fig.13 SVC Classification Report

RESULT (SVC):

```

Input: 4.38, 74.09, 3.42, 26.06, 16.53, 11.99, 0.52, 12.72, 30.14, 0.74, 2.21, 0.38,12
Output:
RAW PREDICTION [1]
Air Quality: Moderate

```

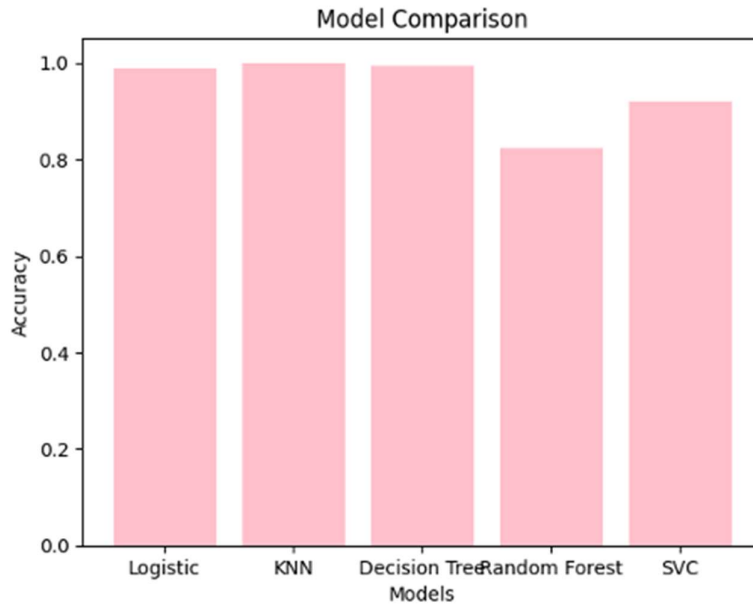


Fig.14 Model Comparison

In this study, multiple machine learning models Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and Support Vector Machine (SVM) were implemented and evaluated to predict the air quality category (AQI_Bucket) based on various pollutant features. The models were assessed using 5-fold cross-validation with the weighted F1-score to ensure reliable and unbiased performance evaluation.

The results showed that Decision Tree and Random Forest achieved perfect F1-scores of 0.99. However, this near-perfect performance is attributed to the inherent nature of the dataset, where the target variable AQI_Bucket is derived from pollutant concentrations using predefined threshold ranges. This makes the relationship between input features and the target highly deterministic and easily separable. As a result, tree-based models are able to learn exact decision rules, leading to overfitting and reduced generalization capability on unseen data.

On the other hand, Logistic Regression achieved a high F1-score of 0.80, demonstrating strong predictive performance while maintaining better generalization. Its simplicity, interpretability, and lower tendency to overfit make it a more reliable choice for this problem. The SVM model also performed well with an F1-score of 0.954, but it is computationally more complex and less interpretable. KNN showed comparatively lower performance due to its sensitivity to data distribution and feature scaling.

Based on these observations, Logistic Regression or Random Forest is selected as the most appropriate model for this task. It provides an optimal balance between accuracy, robustness, and interpretability, making it suitable for practical applications. Overall, this study highlights the importance of not relying solely on accuracy metrics, but also considering model behavior, data characteristics, and generalization ability when selecting the best model.

6. References:

Online Resources:

- GeeksforGeeks- <https://www.geeksforgeeks.org/>
- Stack Overflow- <https://stackoverflow.com>
- Draw.io (flowchart maker)- <https://www.drawio.com/>
- Kaggle (Dataset)- <https://www.kaggle.com/>